

Task-1

Report 1 – Red Team Report

HawkEye Lab – Adversary Simulation Report

Executive Summary

The objective of the attack is to simulate a threat actor using the HawkEye information-stealing malware to compromise a victim workstation, collect credentials, and exfiltrate sensitive information.

Attack Objectives

- Gain initial execution
- Establish outbound communication
- Harvest credentials
- Collect system information
- Exfiltrate stolen data
- Maintain low visibility

MITRE ATT&CK Mapping

Phase	Technique	MITRE ID
Initial Access	User Execution	T1204
Execution	Malicious File	T1204.002
Discovery	System Information Discovery	T1082
Credential Access	Credentials from Web Browsers	T1555
Collection	Automated Collection	T1119
Command & Control	Application Layer Protocol	T1071
Exfiltration	Exfiltration Over Web Services	T1567
Exfiltration	Exfiltration Over Email	T1048

Attack Chain

Phase 1 – Initial Infection

Victim downloads and executes the HawkEye malware.

Possible delivery methods:

- Phishing email
 - Fake invoice
 - ZIP attachment
 - Cracked software
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Phase 2 – Malware Execution

After execution the malware:

- Creates its process
 - Collects victim information
 - Reads browser databases
 - Extracts stored passwords
 - Collects email credentials
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Phase 3 – Network Communication

Observed protocols include:

- DNS
- HTTP
- SMTP
- TCP

The malware contacts external infrastructure to:

- download resources
 - verify Internet access
 - send stolen information
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Phase 4 – Credential Theft

Targets include:

- Chrome
- Firefox
- Outlook
- FTP clients
- Email clients

Collected artifacts include:

- usernames
 - passwords
 - cookies
 - browser history
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Phase 5 – Data Exfiltration

The malware packages stolen information and transmits it through:

- SMTP
- HTTP POST requests

This minimizes attacker interaction while maximizing automation.

Indicators of Compromise

Host IOCs

- Suspicious executable
- Temporary files
- Credential dump artifacts

Network IOCs

- Unusual DNS requests
- HTTP POST uploads
- SMTP traffic toward attacker-controlled email accounts

Red Team Success Criteria

- ✓ Successful execution
- ✓ Credential harvesting
- ✓ External communication
- ✓ Data exfiltration
- ✓ Minimal user interaction

Report 2 – Blue Team Report

HawkEye Lab – Incident Response Report

Executive Summary

The Blue Team investigated suspicious outbound traffic captured in the supplied PCAP. Analysis confirmed the presence of HawkEye stealer malware performing credential theft and exfiltration. The investigation reconstructed the complete attack lifecycle from infection to data theft.

Investigation Workflow

Step 1 – Review PCAP

Tool:

- Wireshark

Identify:

- Top talkers
 - Protocol distribution
 - External IPs
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Step 2 – DNS Analysis

Filter:

dns

Look for:

- suspicious domains
 - newly observed domains
 - repeated lookups
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Step 3 – HTTP Analysis

Filter:

http

Review:

- GET requests
 - POST uploads
 - downloaded payloads
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Step 4 – SMTP Analysis

Filter:

smtp

Recover:

- sender
- recipient
- credentials
- attachments

SMTP traffic is particularly important because HawkEye commonly exfiltrates stolen credentials via email.

Step 5 – Follow TCP Streams

Recover:

- commands
 - payloads
 - stolen credentials
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Step 6 – Decode Data

Use:

CyberChef

Decode:

- Base64
- URL Encoding
- Hex

Findings

Malware Behavior

- Information stealing
- Browser credential theft
- Email credential theft
- Automated exfiltration

Network Findings

Protocols observed:

- DNS
- HTTP
- SMTP
- TCP

Indicators of Compromise

Domains

- Suspicious DNS lookups

IPs

- External C2 servers

Artifacts

- Email credentials
- Browser credentials
- Uploaded data

Detection Opportunities

SIEM Rules

Alert on:

- unusual SMTP clients
 - HTTP POST to unknown domains
 - multiple DNS lookups
 - credential harvesting behavior
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Endpoint Detection

Monitor:

- browser database access
 - credential manager access
 - suspicious child processes
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Network Detection

Deploy IDS signatures for:

- HawkEye
 - SMTP credential theft
 - abnormal HTTP uploads
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Containment

- Disconnect infected host
 - Block malicious domains
 - Block attacker IPs
 - Reset compromised passwords
 - Reimage affected endpoint
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Lessons Learned

Improve:

- Email filtering
- User awareness

- EDR deployment
 - Network monitoring
 - IOC sharing
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Report 3 – Purple Team Report

HawkEye Lab – Attack & Defense Handover Report

Executive Summary

The Purple Team exercise combined offensive simulation with defensive validation to measure the organization's ability to detect and respond to HawkEye stealer malware.

Objectives

Red Team

- Simulate real attacker behavior

Blue Team

- Detect attacker activity
 - Contain compromise
 - Improve detections
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Attack Timeline

Stage	Red Team Activity	Blue Team Response
Initial Access	Execute malware	Process monitoring
Discovery	Gather system info	EDR telemetry
Credential Theft	Browser password theft	Detect browser access
Network Activity	DNS + HTTP	Network monitoring
SMTP	Credential exfiltration	SMTP alerting
Exfiltration	Send stolen data	IOC detection

ATT&CK Coverage

Technique Detection

T1204 Email gateway

Technique Detection

T1082	Endpoint logging
T1555	EDR
T1071	IDS
T1048	SMTP monitoring
T1567	Proxy logs

Detection Gaps

Observed weaknesses:

- Limited SMTP inspection
 - Weak outbound monitoring
 - Lack of browser credential access alerts
 - Insufficient anomaly detection
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Recommended Security Improvements

Email Security

- Attachment sandboxing
 - SPF
 - DKIM
 - DMARC
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Endpoint Security

Deploy:

- Microsoft Defender
- CrowdStrike
- SentinelOne
- Elastic Defend

Network Security

Deploy:

- Suricata
- Zeek
- Snort

Monitor:

- SMTP
- DNS
- HTTP POST traffic

SIEM Improvements

Create detections for:

- Credential dumping
- Browser credential access
- Multiple outbound SMTP sessions
- Data exfiltration

Incident Response

Update playbooks for:

- Credential theft
- Information stealers
- Data exfiltration
- Password reset procedures

Purple Team Outcome

Offensive Success

- Malware executed successfully.

- Credentials were harvested.
- Sensitive data was prepared for exfiltration.
- Outbound communication over DNS, HTTP, and SMTP was established.

Defensive Success

- Malicious traffic was identified in the PCAP.
- Attack stages were reconstructed.
- Network and host IOCs were collected.
- Detection opportunities and response improvements were documented.

Key Recommendations

- Strengthen email security and endpoint detection.
- Monitor outbound DNS, HTTP POST, and SMTP traffic.
- Create SIEM rules mapped to MITRE ATT&CK techniques.
- Conduct recurring Purple Team exercises to validate controls against information-stealing malware.

Useful Links

- CyberDefenders HawkEye Walkthrough: [CyberDefenders HawkEye Walkthrough](#)
- [MITRE ATT&CK Framework](#)
- [Wireshark Documentation](#)
- [CyberChef](#)
- [Sigma Detection Rules](#)